ANDREA LEANG

Cambridge, MA 02139 | 408-460-0551 | akleang@mit.edu | LinkedIn | Website

EDUCATION

Massachusetts Institute of Technology Cambridge, MA

Candidate for MEng. Electrical Engineering and Computer Science *Expected Graduation: May 2026*

S.B. Electrical Engineering and Computer Science *Class of 2025*

Minor in Business Analytics. GPA: 4.8/5.0

Relevant Coursework with Projects: Deep Learning, Multi-agent Learning, Micro/Nano Processing Technology, Engineering for Impact, Design & Rapid Prototyping, Large-scale Symbolic Systems, Robotic Manipulation

Organizations: Gordon-MIT Engineering Leadership Program (GEL), Eta Kappa Nu (HKN), NCAA Women's Fencing, Women's Independent Living Group (WILG), MIT Anime Club, MIT EE Club (Voltage), Random Hall

RESEARCH

MIT-IBM Watson AI Lab | *Master's Thesis Research* Sept. 2025 – Present

- MIT Energy-Efficient Circuits and Systems Group: CPU and GPU-CPU Modeling in the context of large LLM inference, focused on latency, throughput, and accuracy tradeoff in KV Cache and Model Parameter offloading and scheduling

MIT Media Lab | *Undergraduate Researcher in Responsive Environments Group* June 2024 – Aug 2024

- Networked Electronic Textile Skin for VR/AR: Developed an Arduino and Unity system with length, slope, and object detection features to create a real-time 3D digital twin of a grid of 25 sensors embedded in stretchable fabric
- Designed robotics system: derived the digital fabric's position from physical, improving virtual display accuracy
- Engineered bi-directional data communication between Unity and Arduino for real-time control of robotic arms

MIT LEMI | *Co-author, Undergraduate Researcher in Towards Microbial-Arduino Study* June 2023 – June 2024

- Automated transforming bacteria videos to statistics and animations with MATLAB, facilitating data analysis
- Implemented ML clustering techniques to characterize bacterial surface charge, improving data analysis accuracy

EXPERIENCE

Cambridge Mobile Telematics June 2025 – Aug. 2025

Software Engineer Intern - Full-Stack *Cambridge, MA*

- Automated RoadClub users' hard-brake events processing with PostgresSQL, Python, & AWS, improving data analysis
- 1st place company-wide hackathon, integrated virtual pets into DriveWell app with React.js, Node.js, and PostgresSQL
- Developed 2 API handlers and unit tests in Python to enhance date-specific hard brake summary AWS S3 retrievals
- Implemented API calls and UI for hard brake information in Swift, creating 1 feature with 5 layers of iOS app structure

Fencing Star U.S. Distribution Aug. 2023 – Present

Co-Founder and CEO *Remote*

- Distributed 2000 pairs of fencing shoes, generating $100K+ ARR, $180K+ sales, 10K searches since July 2024 launch
- Directed over 25 fencing clubs and NCAA fencing teams (UCSD, Yale, etc.), 5 Team USA, and 1 Olympian partnerships
- Led operations & logistics in 200+ ads, overseas supply chain, and 12 national competition ventures for 5000+ fencers

EXTRACIRRICULARS

MIT Varsity Fencing | *4-time NCAA Regionals Qualifier, Pan-American Finalist* Sept. 2021 – Present

- Squad Leader '22-'25: Led team through competitions and practices 10 hrs/wk against 21 NCAA teams
- US Fencing Coach Association (USFCA): Scholar of Distinction ('22, '23, '24, '25)

Mathematics for Computer Science / Intro to Algorithms | *Teaching Assistant* Feb. 2024 – Present

- Taught probability, graph theory, and proof-writing skills 20 hr/wk to 1200+ undergraduate and graduate students
- Led review sessions and 1-1 tutoring. Turned foundational concepts into digestible content for students of all levels

SKILLS

Technical Skills: Python, C++, MATLAB, LLM and ML Algorithm Design, Nano-fabrication, Photolithography, DRIE, Robotics Simulation, PCB Design, System Diagramming, EDP Simulation, Hardware Attacks & Defenses, USACO Gold